API and Data Sources

Free open-data integrations, endpoint contracts, usage limits and shade caveats

No API keys in this build. These public services are appropriate for a low-volume, non-commercial demo when their current policies are followed. They are best-effort services, not production SLAs.

1. External services

Service	Use	Policy / operating note
OpenStreetMap tiles	Map display in Leaflet	Visible attribution is on the map. Standard tile servers have no SLA; obey caching and no-prefetch rules.
Overpass API	Nearby walkable street geometry, trees and building footprints	Small bounding box only; requests cached 15 minutes. Shared public instances can be busy or rate-limited.
Open-Meteo	Hourly temperature, humidity, clouds and shortwave radiation	Free API is limited and non-commercial; CC BY 4.0 attribution applies.
Local routing	Shortest and heat-aware walk on downloaded OSM street graph	No paid route provider and no external routing quota; quality depends on OSM pedestrian tags.

2. Shade estimate and limits

Open street data does not give a dependable, time-specific shade percentage for every footpath. This implementation checks for mapped trees near a segment and nearby building footprints, then assigns a clearly labeled proxy value. It does not calculate the direction and length of each building shadow, tree canopy size, seasonal foliage, or the user's side of the street. Treat the result as a route-comparison hint and validate it on foot before relying on it.

3. CoolRoute API

Endpoint	Request / response
GET /api/v1/health	Local service status and data mode
GET /api/v1/real/area?bbox=s,w,n,e	Street graph, street geometry, mapped trees, source and attribution. Window capped to a few kilometres.
GET /api/v1/real/weather?lat=...&lon;=...&hour;=...	Hourly forecast values and source/licence metadata.
POST /api/v1/real/route	origin/destination lat/lon, bbox, hour, heat_budget_pct, step_free, shade_available, closed_edge_ids; returns both routes, metrics, weather and warnings.
GET /api/v1/graph, /sun; POST /api/v1/route	Retained synthetic demo endpoints for reproducible offline engine examples.

4. Important public-service policies

- OSMF tile policy: visible attribution; respect cache headers; no bulk tile download or prefetch.
- Nominatim policy: public Nominatim is deliberately not included as a generic search box; use map clicks or browser location in this demo.
- Open-Meteo terms: free API is non-commercial, rate-limited and licensed with CC BY 4.0; commercial integrations need an appropriate subscription.
- Overpass commons guidance: shared public endpoints have capacity limits; keep queries small and cache results.
- OSM copyright and licence: retain clear OpenStreetMap contributor attribution.